

Olist Delivery Reliability and Customer Reviews

SQL and Power BI case study

Brazilian E-Commerce Public Dataset by Olist, 2016–2018

Main question

Which delivery routes produced more late orders than expected, and how were late deliveries related to customer reviews?

Summary

Late orders received much worse reviews than on-time orders. A small number of interstate routes produced most late deliveries. After month and distance were taken into account, SP → RJ remained the largest problem route. On the priority routes, most of the additional time appeared after carrier handoff.

1. Data and method

The source contains orders, customers, order items, sellers, payments, reviews, products and geolocation data. The analysis was built at order level.

Measure	Value
Delivered orders in the main analysis	96,470
Orders with reviews	95,824
Late orders	7,826
Late delivery rate	8.11%
Bad review rate	12.81%
Single-seller interstate orders	60,961
Interstate orders with distance	60,551 (99.33%)

- A bad review means a score of 1 or 2.
- Order items and payments were aggregated before joining to orders.
- The latest review row was kept for each order.
- Route analysis uses single-seller orders so that each order has one origin state.
- Distance is the straight-line distance between average ZIP-prefix coordinates.

2. Late delivery and customer reviews

Delivery group	Orders	Reviewed orders	Average score	Bad review rate
On time or early	88,644	88,163	4.29	9.22%
Late	7,826	7,661	2.57	54.07%

A bad review was 5.86 times more common among late orders. The average score fell from 4.29 to 2.57.

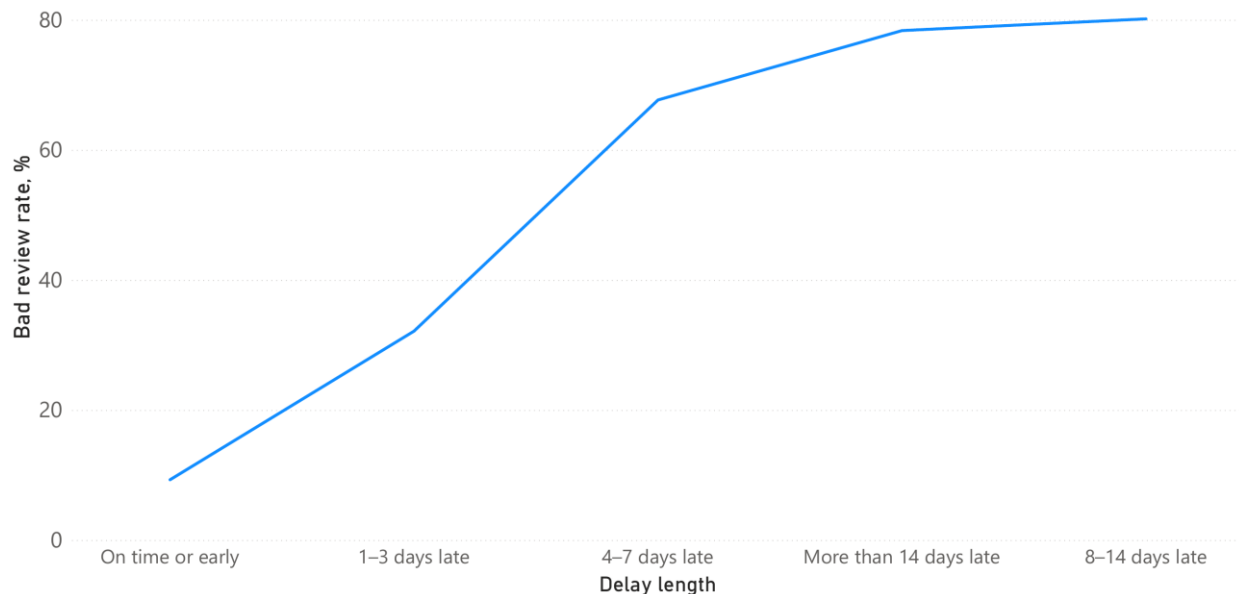
Delivery Reliability Executive Overview				
96470 Analyzed orders		6534 Late orders		62,42 % Bad review rate among late orders
				2,27 Average review score among late orders
Route	Adjusted excess late orders	Actual late rate, %		Expected late rate, %
SP → RJ	689,40	14,25		5,65
SP → BA	104,40	12,89		8,28
SP → ES	75,40	12,33		7,07
SP → SC	48,70	9,57		7,45
PR → RJ	39,00	12,34		8,22

3. Delay length

Delay group	Reviewed orders	Average score	Bad review rate
On time or early	88,163	4.29	9.22%
Less than 1 day late	1,280	4.03	12.42%
1–3 days late	1,852	3.29	32.13%
4–7 days late	1,748	2.10	67.68%
8–14 days late	1,446	1.67	80.15%
More than 14 days late	1,335	1.72	78.35%

The result changed sharply after about four days. For orders delayed by 4–7 days, 67.68% of reviews were bad.

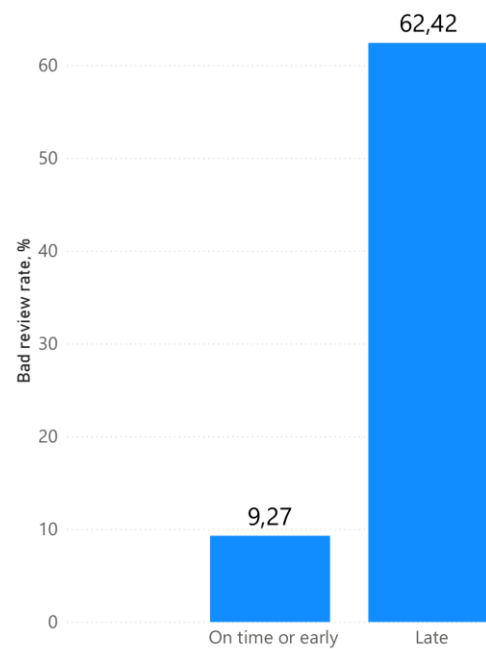
Bad-review rate rises sharply after four days of delay



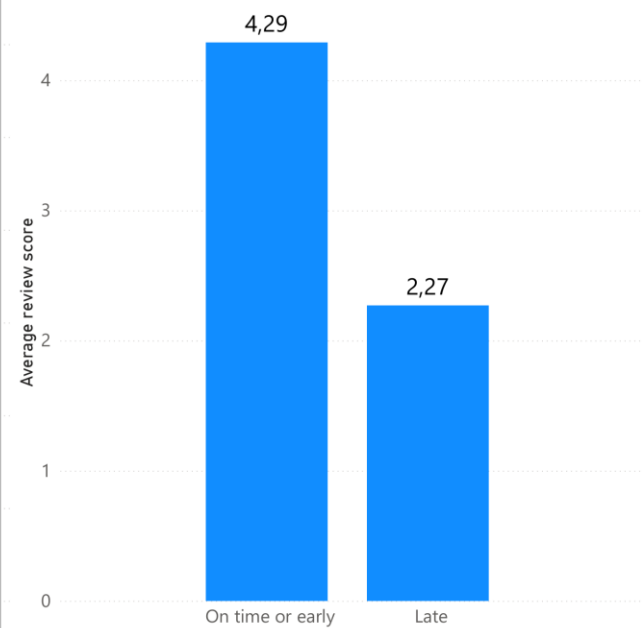
4. Route concentration

There were 391 interstate route pairs. Twenty-four routes, or 6.14% of all route pairs, produced at least 80% of interstate late orders. SP → RJ alone produced about 22% of interstate late orders.

Bad reviews are 6.73* more common when delivery is late



Average review score falls from 4.29 to 2.27 when delivery is late

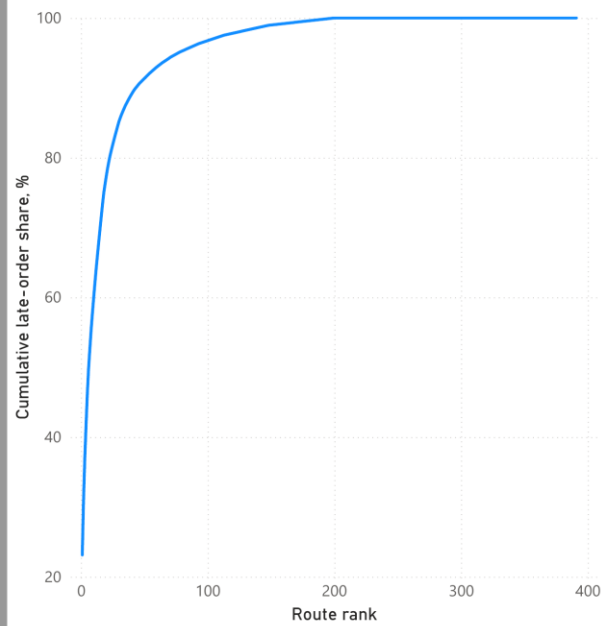


5. Distance and late delivery

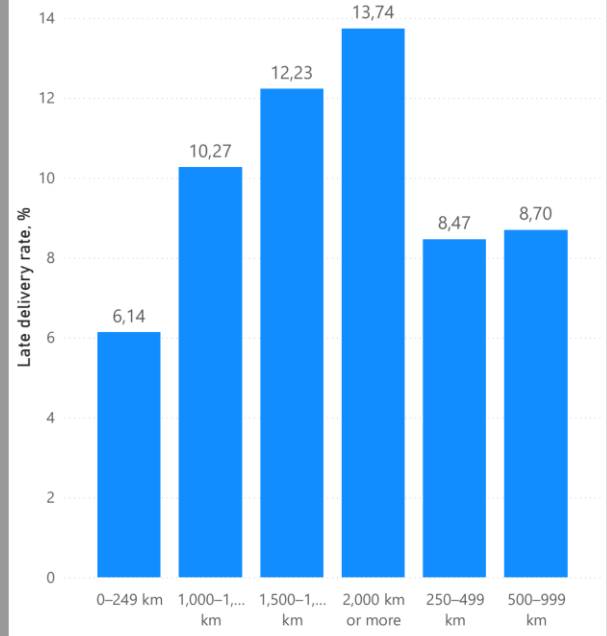
Distance	Orders	Late rate	Average delivery	Median carrier time
0–249 km	1,970	6.14%	11.68 days	172 hours
250–499 km	18,861	8.47%	12.75 days	175 hours
500–999 km	24,528	8.70%	14.04 days	214 hours
1,000–1,499 km	6,487	10.27%	16.88 days	285 hours
1,500–1,999 km	3,132	12.23%	19.01 days	330 hours
2,000 km or more	5,573	13.74%	20.75 days	355 hours

Longer deliveries were more likely to be late. Route performance was therefore compared with other routes in the same month and distance band.

24 of 391 routes account for 80% of interstate late orders



Late rate rises from 6.14% to 13.74% with distance

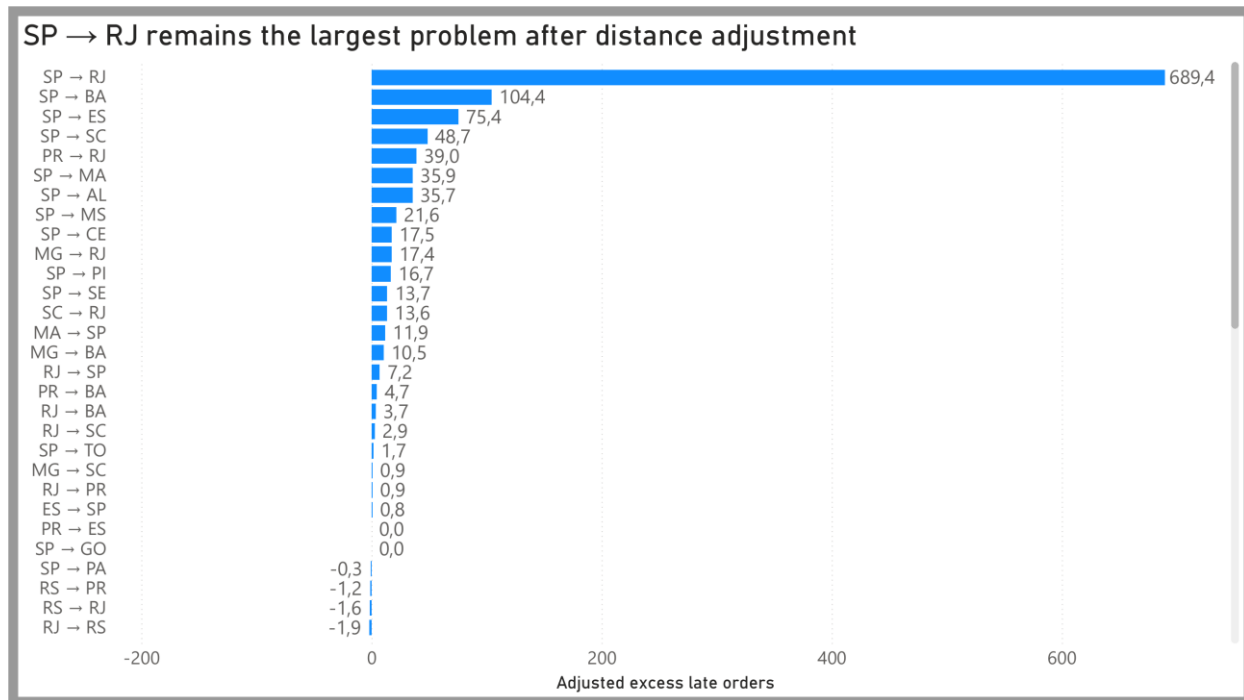


6. Routes with more late orders than expected

Expected late orders were calculated from other routes in the same month and distance band. The route itself was excluded from its benchmark.

Route	Orders	Actual late	Expected late	Excess late	Actual vs expected
SP → RJ	8,021	1,253	535.3	717.7	15.62% vs 6.67%
SP → BA	2,265	340	210.8	129.2	15.01% vs 9.31%
SP → ES	1,436	203	120.3	82.7	14.14% vs 8.38%
SP → SC	2,299	254	198.6	55.4	11.05% vs 8.64%
PR → RJ	948	135	91.3	43.7	14.24% vs 9.63%
SP → MA	482	102	62.2	39.8	21.16% vs 12.90%
SP → AL	254	67	27.8	39.2	26.38% vs 10.93%

SP → RJ remained the largest problem after adjustment. SP → PA, SP → DF and SP → RN no longer looked poor after distance was taken into account.



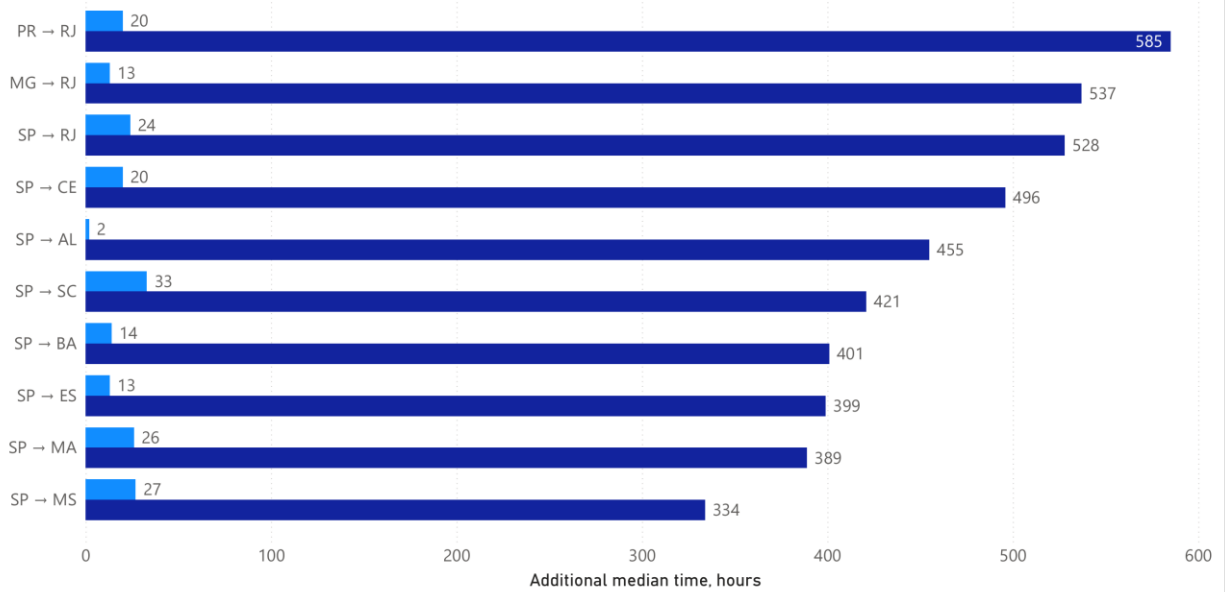
7. Where the delay accumulates

Group	Median time before carrier	Median time after carrier handoff
On-time orders	42 hours	165 hours
Late orders	72 hours	575 hours
Difference	+30 hours	+410 hours

On the ten priority routes, the additional median time before carrier handoff ranged from –1 to 27 hours. The additional time after carrier handoff ranged from 358 to 600 hours.

Most additional time accumulates after carrier handoff

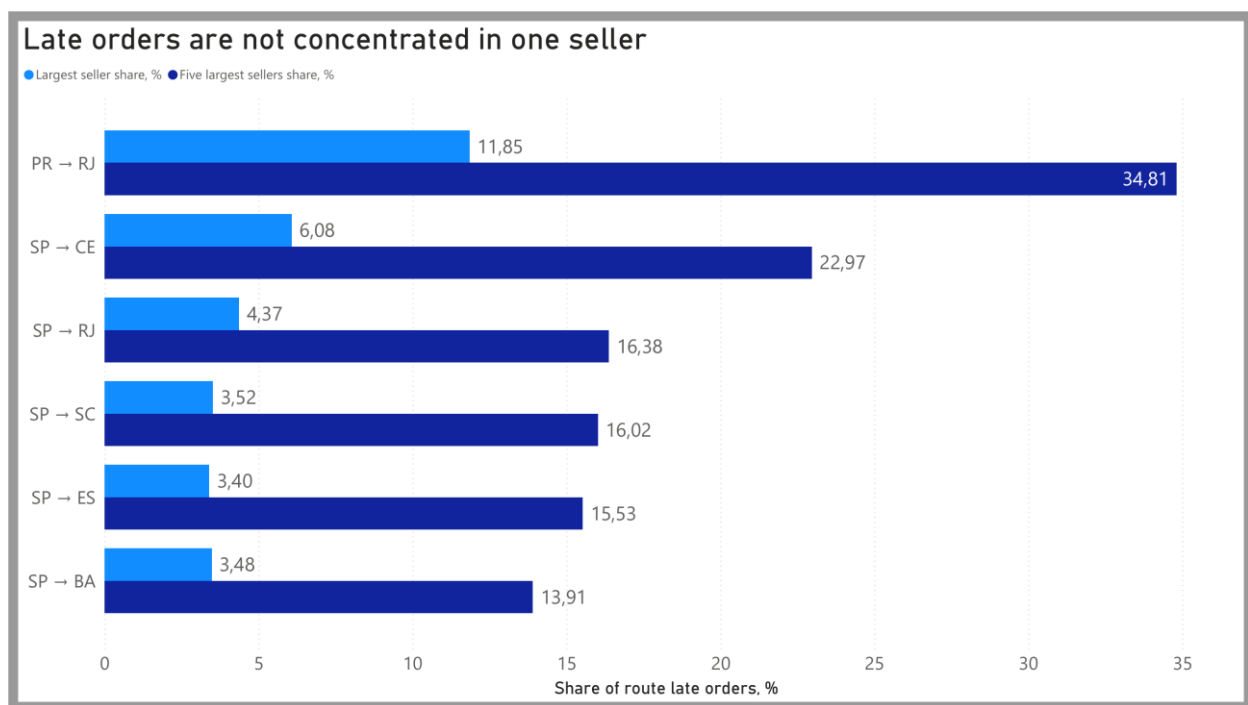
● Before carrier handoff ● After carrier handoff



8. Seller concentration

The problem was not concentrated in one seller. Across the ten priority routes, the largest seller produced 3.40%–11.85% of late orders. The five largest sellers produced 13.91%–34.81%.

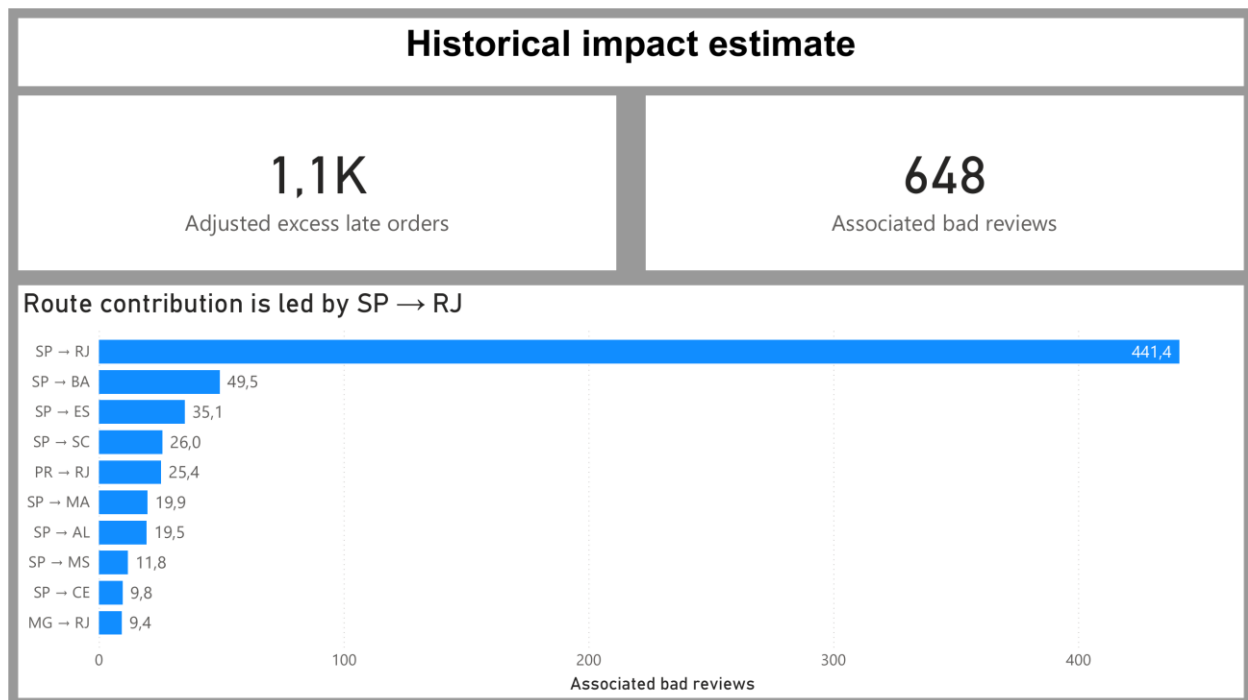
Route	Seller count	Top seller share	Top five sellers share
SP → RJ	1,002	4.37%	16.38%
SP → BA	594	3.48%	13.91%
SP → ES	480	3.40%	15.53%
SP → CE	344	6.08%	22.97%
PR → RJ	178	11.85%	34.81%
SP → SC	589	3.52%	16.02%



9. Historical impact estimate

The ten largest adjusted problem routes produced 1,168.5 late orders above their month-and-distance benchmarks. Using the observed difference in bad-review rates between late and on-time orders within each route gives an estimate of about 627 associated bad reviews.

This is a historical comparison, not a guaranteed forecast and not a causal estimate.



10. Recommendations

1. Investigate SP → RJ first. It has 717.7 late orders above its benchmark and an average distance of 441 km.
2. Check carrier handoff, intermediate hubs and service-level performance on SP → RJ, SP → BA and SP → ES.
3. Track actual late rate, expected late rate, excess late orders, order volume and bad review rate by route each month.
4. Contact customers proactively when an order approaches four days late.
5. Use excess late-order count to set priorities. Use excess percentage points to find smaller high-risk routes.
6. Test changes on a small number of routes and compare them with similar routes before making causal claims.

11. Limitations

- The data covers Olist orders from 2016–2018.
- The analysis is observational and does not prove causation.
- State-to-state routes are broad route groups, not exact carrier lanes.
- Distance is straight-line distance between ZIP-prefix centroids.
- Multi-seller orders are excluded from route analysis.
- Carrier identity is not available.
- Late delivery was defined from the source delivery timestamps.